Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Ligler, Frances Ph.D., D.Sc.		
Project Title:	Multianalyte Biosensor with Chips for Human and Environmental Monitoring		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Fligler@cbmse.nrl.navy.mil	**Fax:**	FY 202-404-8897
PI Organization Type:	GOVERNMENT	**Phone:**	202-404-6002
Organization Name:	Naval Research Laboratory		
PI Address 1:	Center for Bio/Molecular Science and Engineering		
PI Address 2:	Code 6900		
PI Web Page:			
City:	Washington	**State:**	DC
Zip Code:	20375-5348	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-01
Start Date:	02/09/2000	**End Date:**	02/08/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Golden, Joel (National Research Laboratory) Rowe Taitt, Chris (National Research Laboratory)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The closed environments of space travel offer enhanced opportunities for transmission of pathogens and toxins. Ideally, food, water, air, and people need to be monitored in order to assure the continued health of the traveler. Rapid diagnostic systems are required, in particular, in cases of human illness to identify both the infection and the source. We propose to build a handheld, low-power biosensor capable of analyzing samples from virtually any aqueous medium for a customized array of targets. The sample will be exposed to a sensing chip with an array of specific (antibody) and semi-selective binding molecules capable of recognizing multiple targets. The sample is followed by a fluorescent reagent capable of binding to any captured target. The identity of the target is determined by the location of the fluorescence on the chip. In the first year, the biosensor itself will be miniaturized, while the second year will focus on the design of chips that can be mass produced and include both a wave guide coated with the capture array and the		

	reservoir of fluorescent reagent. In the third year, chips will be adapted specifically for monitoring: (1) food and water, (2) air and surface samples collected using a small cyclone or swipe, respectively, and (3) human fluids such as nasal swabs and blood. Targets will include bacteria, viruses, toxins and metabolic indicators.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This same sensor can be used for infectious disease diagnosis and rapid identification of biological warfare agents. The United States Marines are supporting a company to develop the array biosensor for the latter application. The Department of Defense (DoD) has just initiated an ancillary project to demonstrate the ability of the array biosensor to monitor food safety and a small company is exploring its use for veterinary diagnostics.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)